



ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA

Internet Data Center and Cloud infrastructures

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Real datacenters



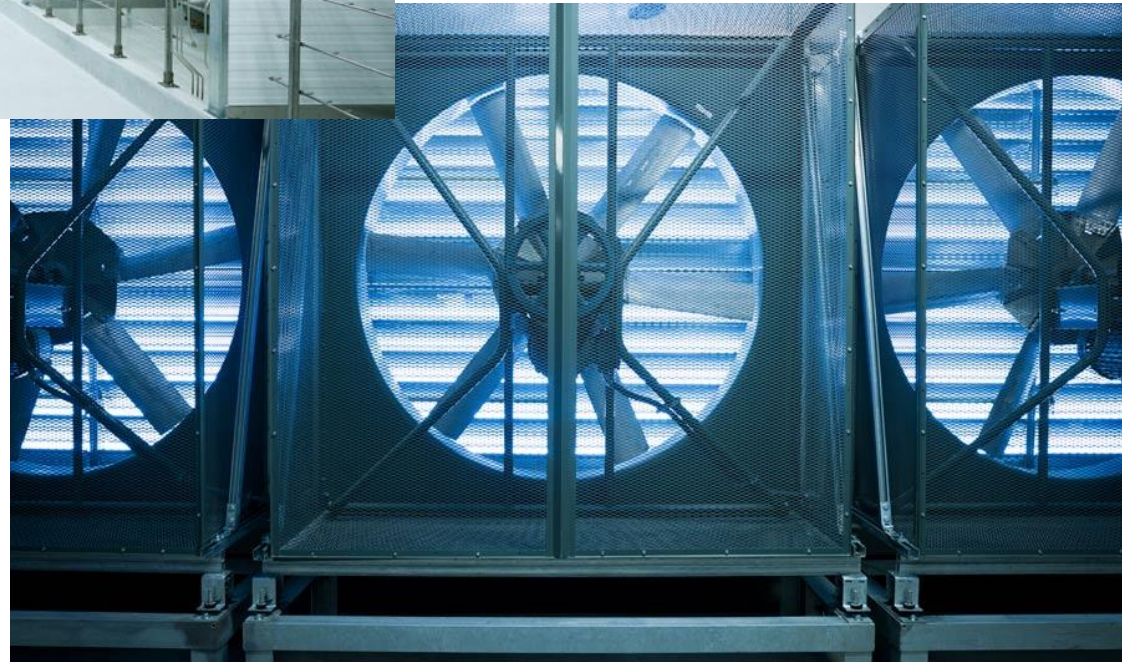
Google datacenter



Facebook datacenter



Facebook datacenter (cooling system)



Microsoft Azure



Apple (one iCloud center)



1 THE NEW QUESTION MARK
Is this the site of Apple's 4.8-megawatt biogas plant?

2 SubStation

3 Apple's 500,000 square foot data center. Home to the iCloud.

4 A smaller 21,000 square foot "tactical" data center.

5 The 100-acre solar farm will generate 20 megawatts of power.



Apple



Italy (~Pavia): ENI datacenter



Italy (Milano): New Aruba datacenter

Global Data Center

- Milan, Italy
- 100% green energy
- 90 MW
- Tier4
- 40.000m² in a 200.000m² campus



Italy (Bologna): ECMWF + Technopole



Some videos

- **Facebook:**

<https://www.youtube.com/watch?v=frzVtaNrHU0>

- **Google:**

<https://www.youtube.com/watch?v=zDAYZU4A3w0>

<https://www.youtube.com/watch?v=kd33UVZhnAA>

- **Microsoft:**

<https://www.youtube.com/watch?v=XgPH9ndCbxs>



An IDC should care not only of IT issues

- Interconnection to Internet (*first mile*)
- Architectures for logical security
- Logistics
- Cooling systems
- Anti-fire plant
- Physical security (access control, intrusion, alarm system)
- Detectors (smoke, water, temperature, humidity, ...)
- Electrical systems and uninterruptible power supply (UPS)
- Disaster recovery plans
- **Site**
- **Size**



Strong commitment on security (1)

1. PEOPLE

- Analysis of employees (past records and way of conduct, including strong logging of their actions)
- Separation of duties
- Security Team: 100+ experts
- Incident Response Teams 24/7

2. PHYSICAL

- Barriers, alarm, 24/7 monitoring, private policy checks, electronic and biometric accesses, CCTV, multiple generators, ...

3. LOGICAL

- Logical network security, host security, transmission level security, database security, AAL
- Continuous monitoring of threats and vulnerabilities
- Security embedded in the entire lifecycle software development



Strong commitment on security (2)

4. ASSESSMENT and AUDITING

- Mitigate risks while complying with legal, statutory, contractual, and internally developed requirements

5. POLICIES and PROCEDURES

- Develop and enforce policies and procedures at all levels
- Design and secure information systems using security domains, defense in-depth and least privilege principles
- Develop and integrate security architecture into business processes (CobiT, ISO27001) and achieve certifications: Safe Harbor, PCI, SAS70, FISMA, CSA, ...
- Conduct employee security awareness training classes
- Perform regular vulnerability assessments and pen-testing (typically by internals)
- External audits



Physical security

All entrances must be controlled

- badge or biometric systems
- prevent keys (except for emergencies)
- ports must be connected to an alarm system that must be activated if left open for over a period of time
- it is preferable to avoid windows

Disaster Recovery: to prevent and mitigate incidents

- smoke detectors and fire
- where possible, anti-flammable materials
- generators UPS at least 48 hours of power where operations must be tested periodically (even monthly)
- Redundant cooling system
- Backup system alternative far from the datacenter (>30Km is the rule)



Control room (h24 or 24/7)



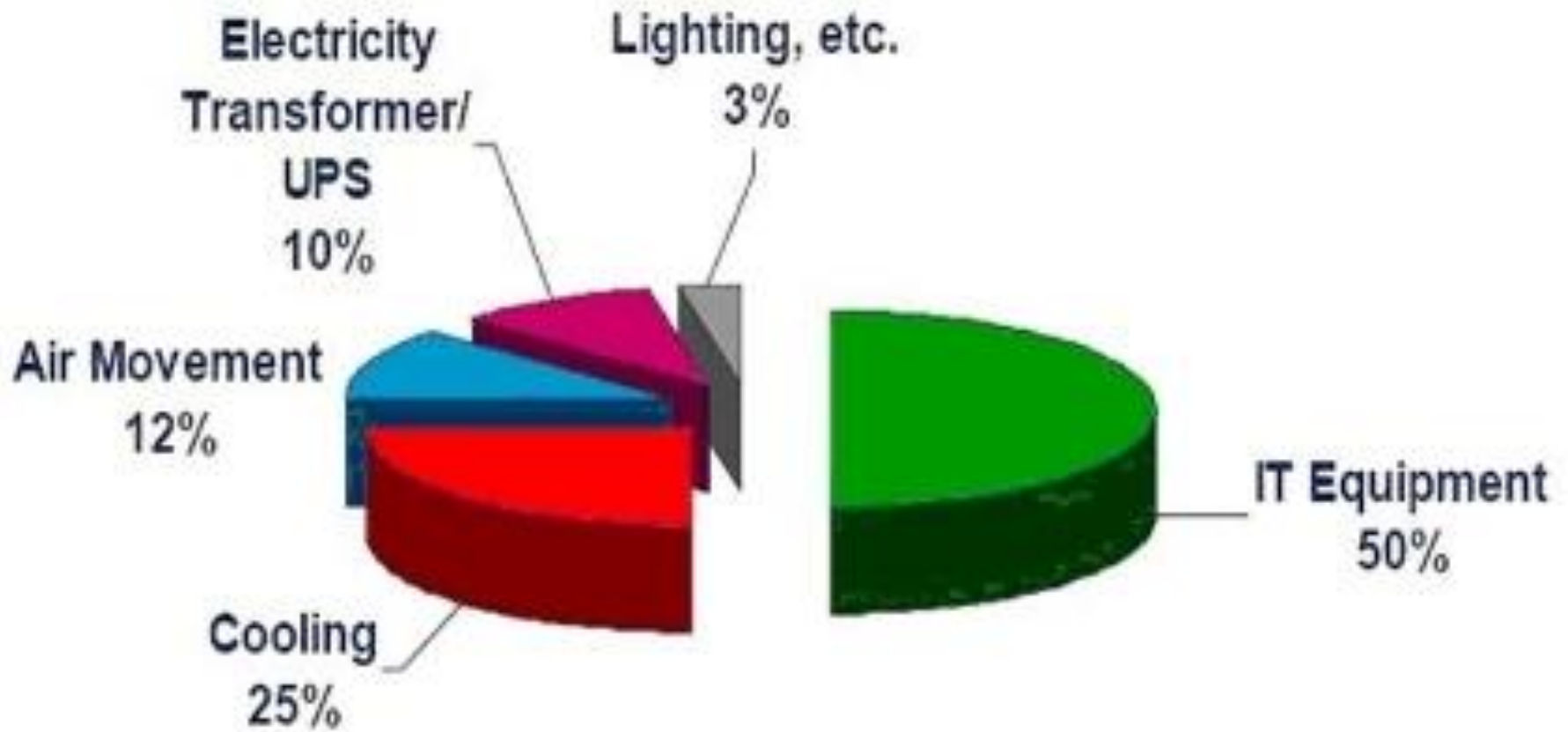
Data Runs the Planet – *Together We Can Ensure it Does Not Ruin the Planet*

Sustainable data centers

- For years, companies that operate large data center infrastructures have been investing in “clean energy” sourcing as part of their strategies for becoming **carbon neutral** or **carbon negative**
- **Google Cloud** was the first major company to go carbon neutral. In 2018 it become the first major company to use 100% renewable energy



Where data centers power goes



Sustainable data centers by design

RESILIENCY

Patented resilient data center designs guarantee 100% Uptime



EFFICIENCY

Efficient design to reduce IT footprint, energy consumption and cost of deployment



ESG

Industry-leader in ESG (Environmental, Social and Governance)



EMISSIONS

Net zero scope 1 & 2 carbon emissions



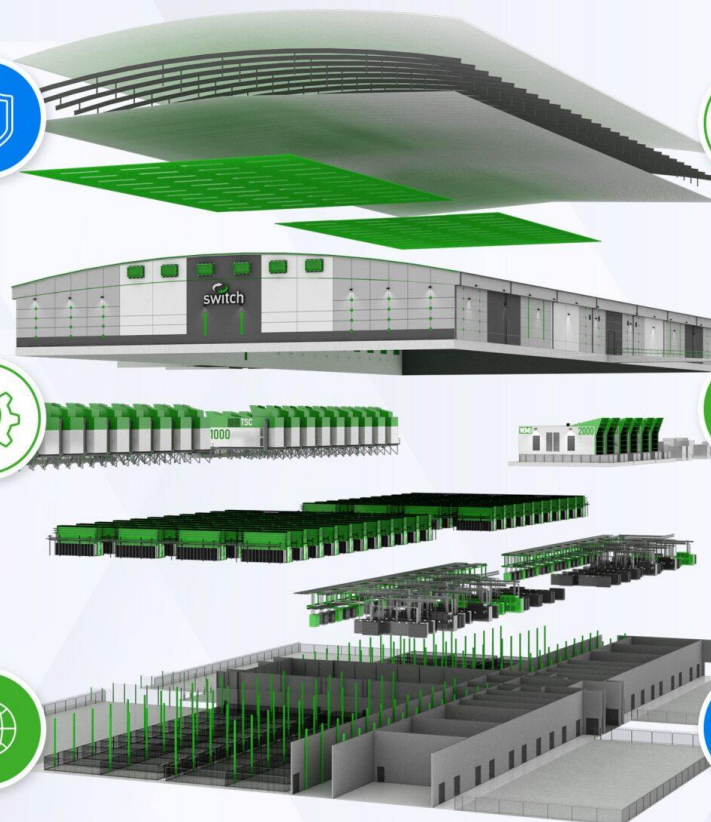
POWER

All Switch data centers have run on 100% renewable energy since 2016

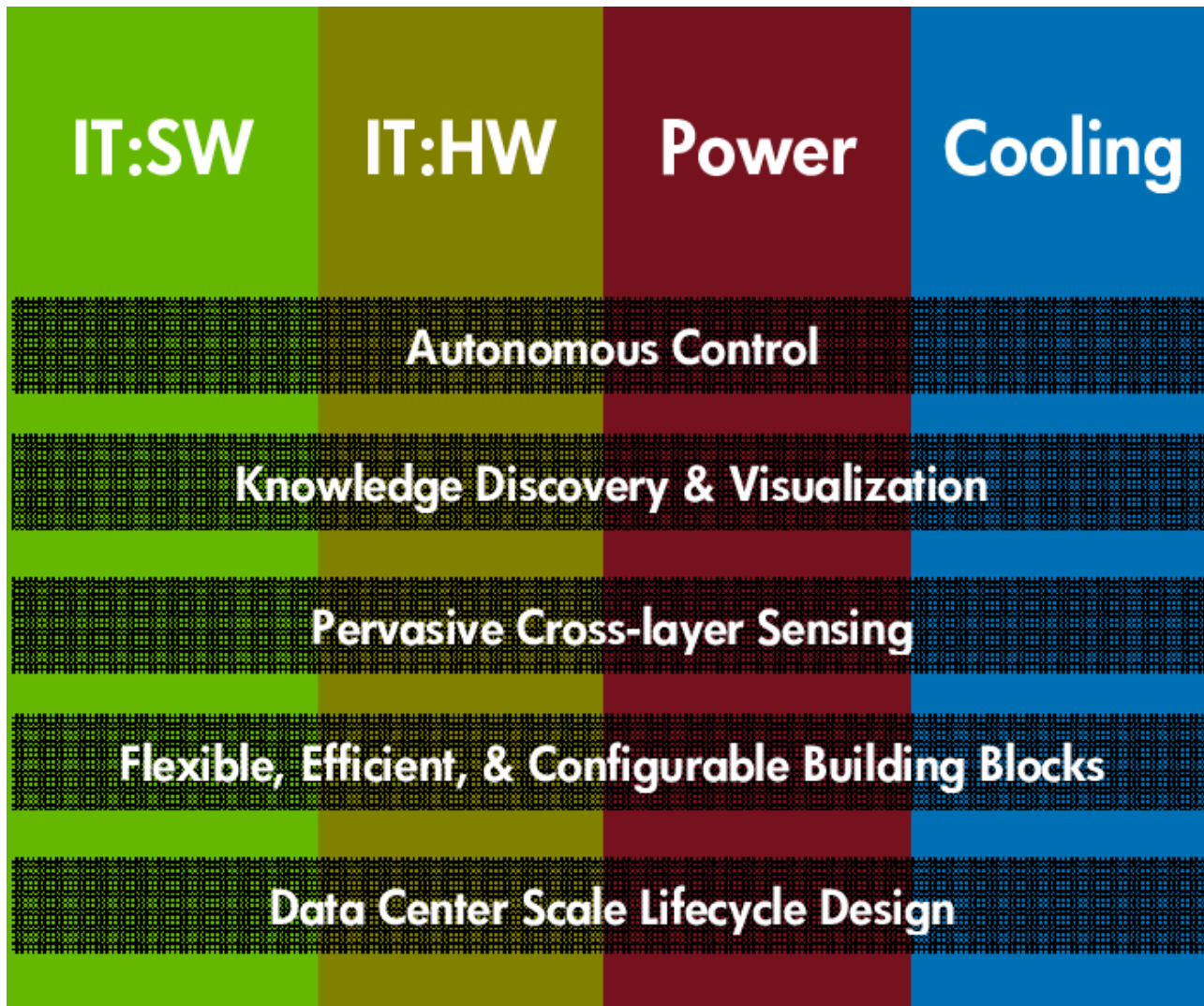


WATER

Switch replenishes 2x its annual water consumption for net positive water impact



Sustainable data centers



Sustainable energy and consumption

Increased investment by tech companies in next-generation technologies that, if they fully pan out, could deliver much more significant energy and carbon savings than resources like solar and wind power

- **Intel** announced an effort to create a reference design for **immersion cooling**, a technique that involves immersing servers and other equipment in non-conductive liquids to enable hyper-efficient cooling
- There is a major interest among tech companies in leveraging **hydrogen** as a power source for data centers. The technology for making this happen on a large scale is not quite there yet
- **Amazon** committed to publishing *Water Usage Effectiveness* reports on a yearly basis. It also pledged to become water-positive, meaning it will make more water available than it consumes
- Microsoft and Google Cloud have made similar pledges to water-positivity in recent years



2nd Trend → Green data centers (*clean energy index*)

-

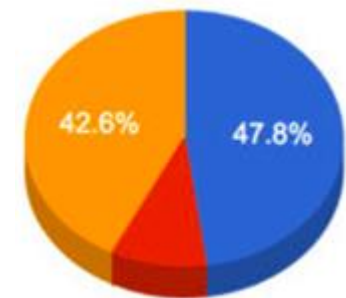
Note: This list is subject to annual variation

Company	Clean Energy Index	Coal	Nuclear	Energy Transparency	Infrastructure Siting	Energy Efficiency & GHG Mitigation	Renewables & Advocacy
	NA	NA		A	C	B	D
	13.5%	33.9%	29.9%	F	F	D	F
	15.3%	55.1%	27.8%	D	F	D	D
	56.3%	20.1%	6.4%	C	C	C	D
	36.4%	39.4%	13.2%	D	B	B	C
	39.4%	28.7%	15.3%	B	C	B	A
	19.4%	49.7%	14.1%	C	D	B	C
	12.1%	49.5%	11.5%	C	D	C	D
	13.9%	39.3%	26%	C	D	C	C
	7.1%	48.7%	17.2%	D	D	C	D
	23.6%	31.6%	22.3%	C	C	C	C
	4%	33.9%	31%	B	C	C	C
	21.3%	35.6%	12.8%	F	D	F	D
	56.4%	20.3%	14.6%	C	B	B	B



Power Usage Effectiveness (PUE) and Data Center infrastructure Efficiency (DCiE)

PUE	DCiE	Level of Efficiency
3.0	33%	Very Inefficient
2.5	40%	Inefficient
2.0	50%	Average
1.5	67%	Efficient
1.2	83%	Very Efficient

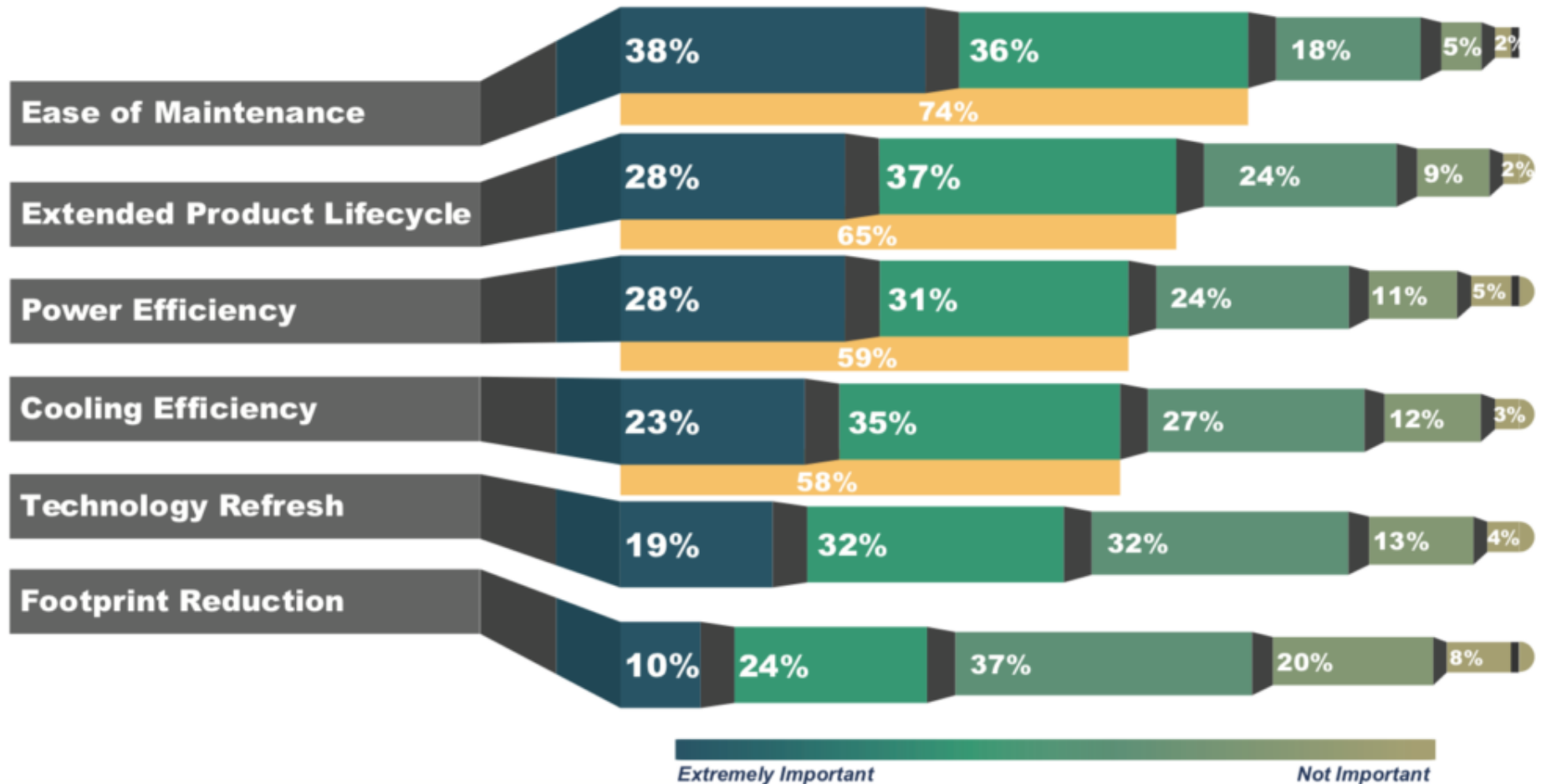


- IT Load (including fans)
- (Of which fan load)
- Cooling Load



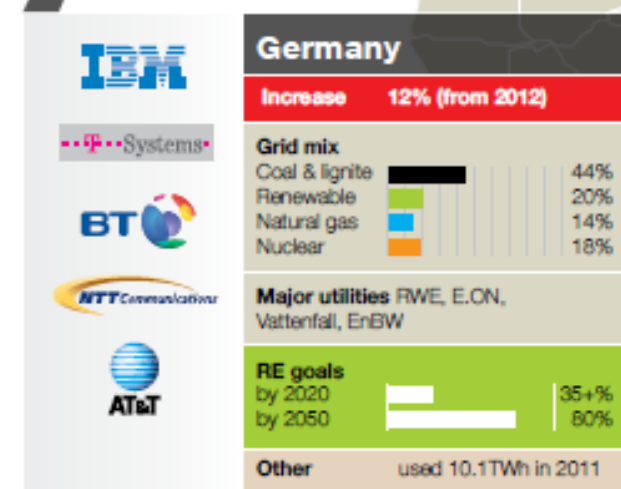
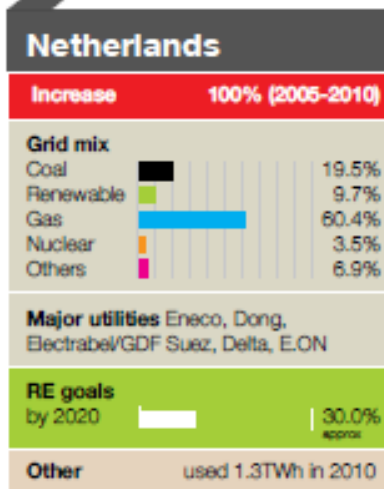
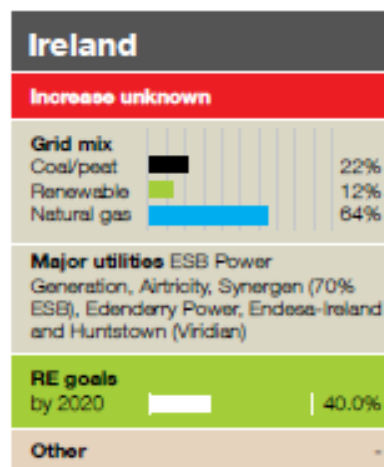
Components for efficiency

Data Center Design Factors



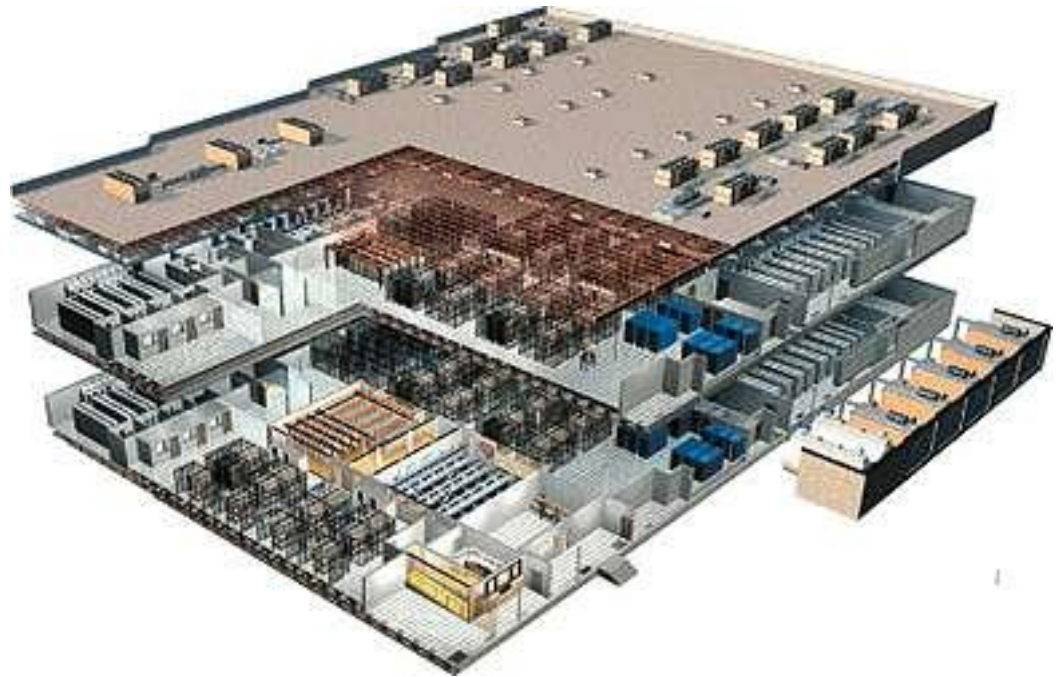
Greener plans

- Key cloud players in the region
- Key utilities and the grid mix
- Renewable energy goals
- Other information

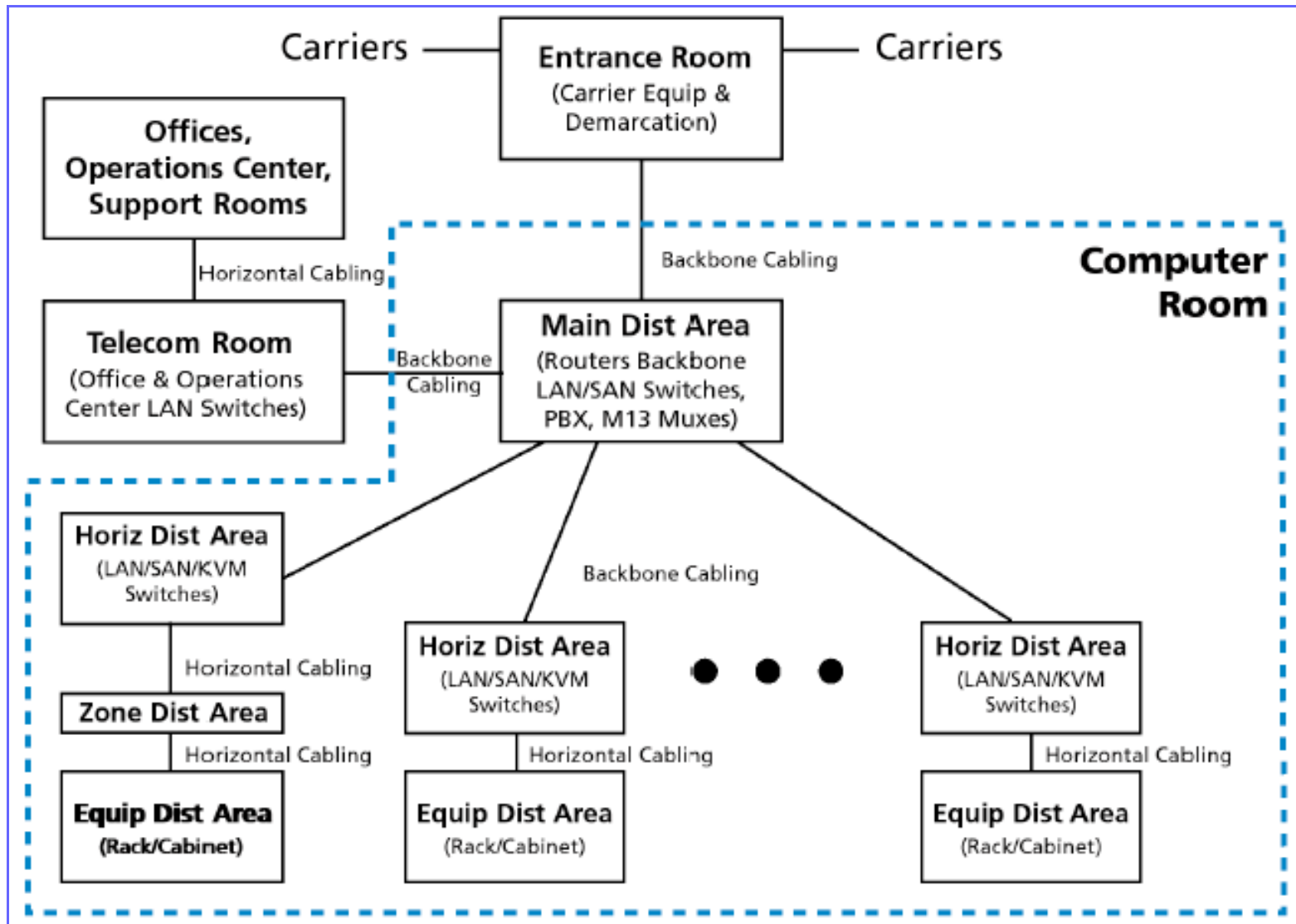


Availability: Standard TIA-942

- **TIA** = *Telecommunications Industry Association*
- **Several data center components**
 - Architecture
 - Networking
 - Power plant
 - Cooling
 - Mechanical parts
 - Security
 - Reliability
 - ...
 - Costs



TIA compliant datacenter (key functional areas)



Tiered reliability for datacenters

Tier-1

- **99,671% availability** (max annual downtime: 28,8 hours)
- *Susceptible to service interruption from both planned and unplanned activity*
- Simple path for cooling and power distribution with no redundant component
- Implementation: 3 months

Tier-2

- **99,741% availability** (max annual downtime: 22 hours)
 - *Simple path for cooling and power distribution with some redundant components*
 - Implementation: from 3 to 6 months
-

Tier-3

- **99,982% availability** (max annual downtime: 1,6 hours)
- *Enables planned activities without interrupting services, but unplanned events may cause disruption*
- Multiple paths for cooling and power distribution with some redundant components
- Implementation: from 12 to 20 months – **Costs: 2 x Tier-1**

Tier-4

- **99,995% availability** (max annual downtime: 15 minutes)
- *Planned activities do not interrupt services, and data center can sustain at least one worst-case unplanned event with no critical impact on services*
- Multiple paths for cooling and power distribution with all redundant components
- Implementation: from 15 to 20 months

Glossary



Definitions (1)

- **Airside economizer** - A mechanical device used to take outside air into the data center when the air temperature is at or below the set cooling point.
- **Colocation (retail)** - Placement of IT equipment in a provider's shared data center space. Targets smaller requirements for floor space and power requirements (typically less than 500 kW for total IT load). Colocation agreements can range from one to five-year terms and may include other items such as caged space or managed services.
- **Container** - Originated from the use of an ISO (International Standards Organization) intermodal shipping container being used to accommodate data center racks. Many variations of the 20 and 40 foot long containers exist and fulfill needs such as quick deployment and mobility. Power and cooling plants are also available in ISO shipping containers.
- **(hot or cold aisle) Containment** - Placing isolation barriers at the front, back and top of aisles to eliminate air mixing.
- **DCIM (Data Center Infrastructure Management)**. The integration of IT and facility management disciplines to centralize monitoring, management and intelligent capacity planning of a data center's critical systems.



Definitions (2)

- **Disaster Recovery** - The process, policies and procedures related to preparing for recovery or continuation of technology infrastructure critical to an organization after a natural or human-induced disaster. Disaster recovery is a component of a Business Continuity Plan (BCP) which focuses on aspects of a business functioning in the midst of disruptive events
- **Evaporative cooling** - Also known as swamp cooling, this is a way to cool air that takes advantage of the drop in temperature that occurs when water that is exposed to moving air begins to vaporize and change to gas
- **HPC - High Performance Computing**. The use of parallel processing for running advanced application programs efficiently, reliably and quickly. The term applies especially to systems that function above a teraflop (10^{12} floating-point operations per second)
- **IRR** - The Internal Rate of Return on an investment project is the annualized effective compounded return rate' or discount rate that makes the net present value (NPV) of all cash flows (both positive and negative) from a particular investment equal to zero
- **ISP** - Internet Service Provider



Definitions (3)

- **ISO/IEC 20000** - The first international standard for an IT Service Management System. It is based on and supersedes BS 15000
- **MSP** - Managed Service Provider
- **MicroGrid** - A localized grouping of electricity generation, energy storage, and loads that normally operates connected to a traditional centralized grid (utility) connected to it - generators, consumers and those that do both – in order to efficiently deliver sustainable, economic and secure electricity supplies
- **Modular** - A conceptual definition for a method of deploying components of, or a complete data center. Modularity implies a pre-packaged, sometimes manufactured module of data center components that through its standardized and pre-defined parameters enable scalability and rapid delivery schedule
- **PUE (Power Usage Effectiveness)** - A metric developed by The Green Grid (thegreengrid.org) to determine the energy efficiency of a data center. PUE is calculated by dividing the amount of power entering a data center by the power used to run the IT load within it
- **REIT (Real Estate Investment Trust)** - A security that sells like a stock on the major exchanges and invests in real estate directly, either through properties or mortgages



Definitions (4)

- **UPS (Uninterruptible Power Supply)** - The UPS is intended to provide clean undisturbed stabilized AC voltage, within strict amplitude and frequency limits, to protect the IT load from any utility power disturbances and irregularities, including outages for a limited time dictated by the capacity of the battery bank. The term UPS refers generally to AC Static systems, other types include DC and Rotary UPS
- **Water Economizer** - A mechanical device used to take cold air and cool an exterior water tower. The water from the chilled tower is then used in the air conditioners inside the data center instead of mechanically chilled water
- **Wholesale data center service** - Leasing of a fully-built data center space. The wholesale tenant gains more control over how their space is used for security and supporting infrastructure. Having the built-out space that is supplied with electrical and environmental infrastructure allows the tenant quick time-to-market in getting their IT infrastructure up and running



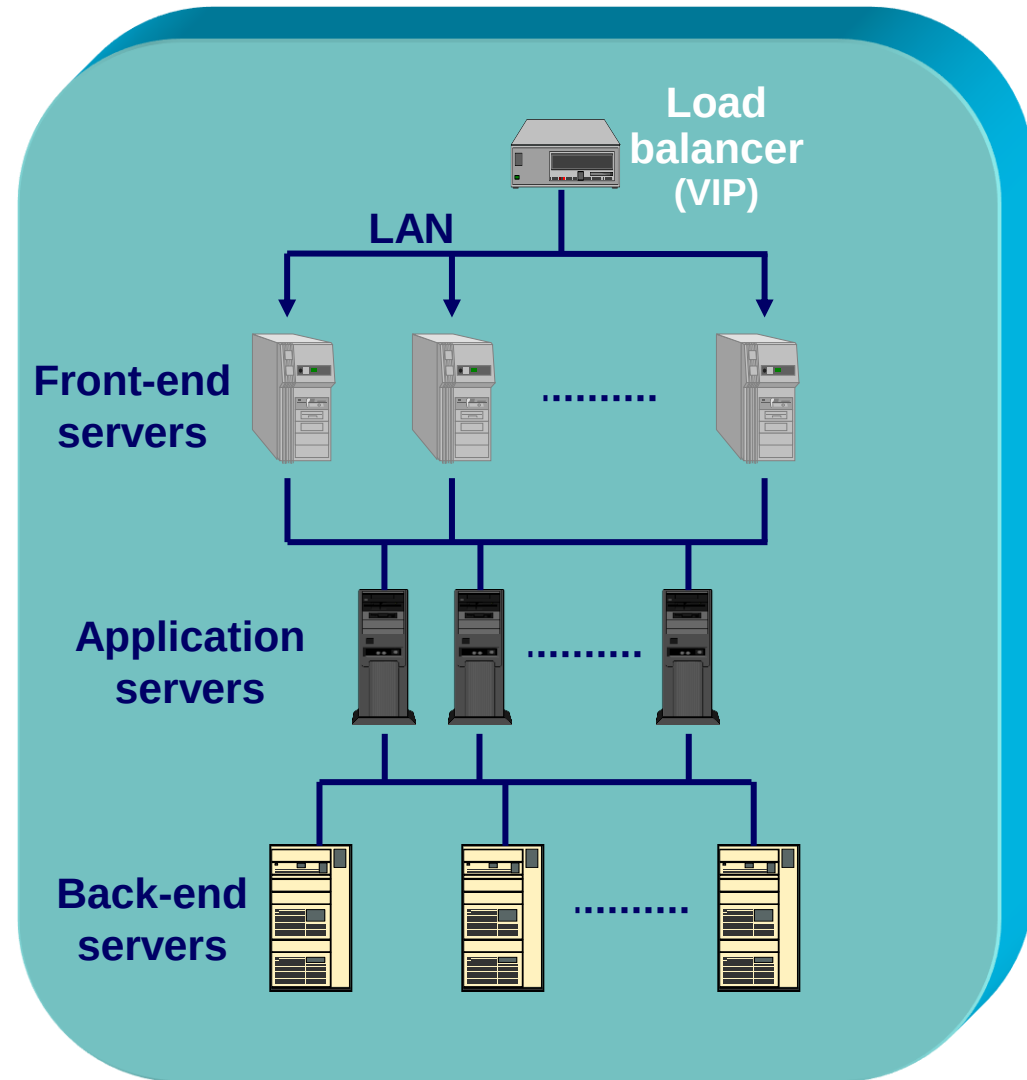
Cloud infrastructures



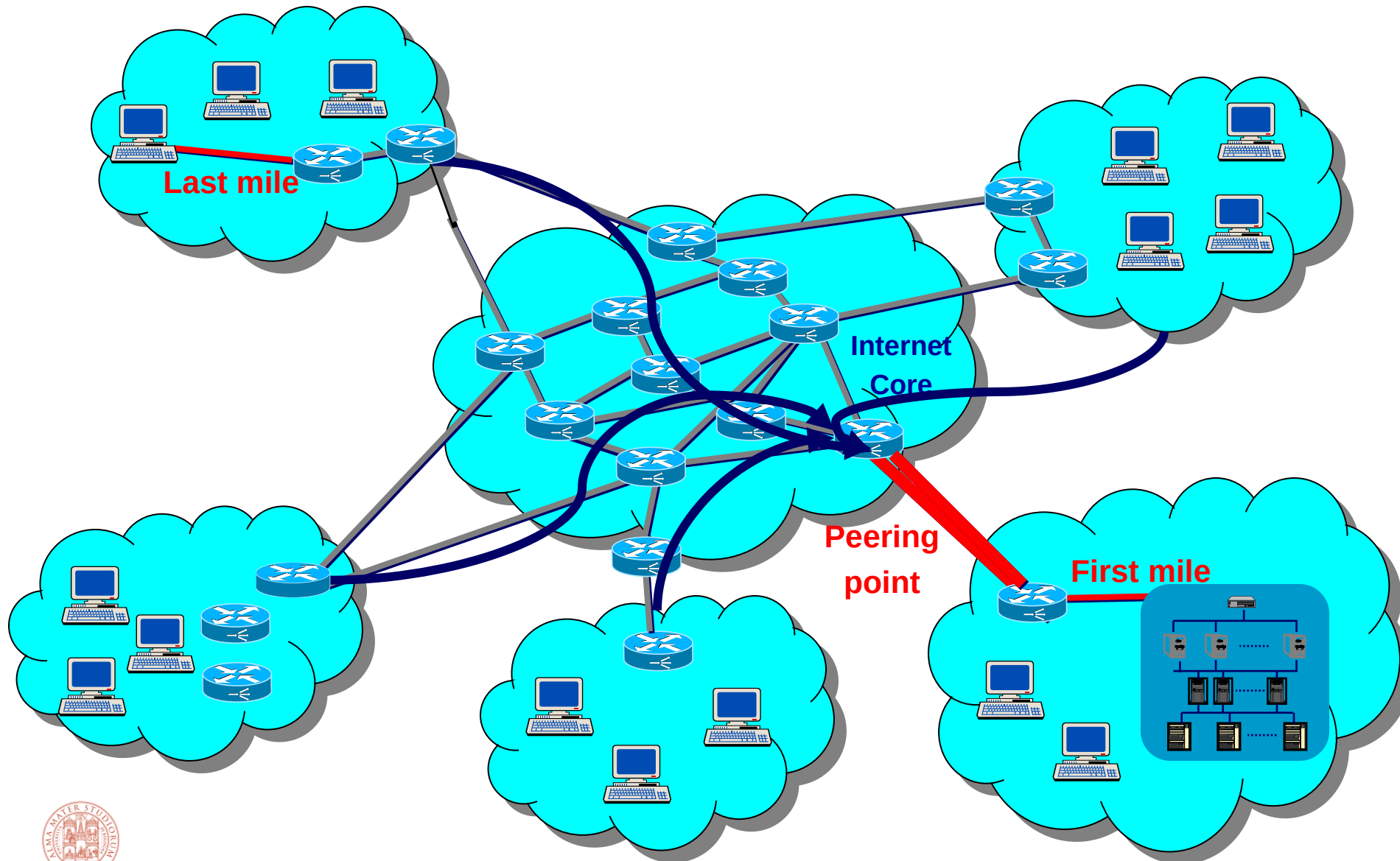
The last question of the previous lesson

1) *How many servers can we add?*

2) *How many servers it is convenient to add?*



One datacenter has evident limits of *reliability* and *communication scalability*



Reliability

If your datacenter is really popular, then you have high risks. But you can afford large investments, hence

- You cannot take any kind of risks about **internal reliability**
 - Infrastructure (remember: from T1 to T4 tier)
 - Employees (faithful, competent, available even at 2 AM?)
- You cannot take any kind of risks about **external reliability**
 - Network: several single points of failure and bottlenecks
 - Electrical power
 - Road works
 - Bad weather, floods, earthquakes, fire, ...
 - Human: cyber-attackers, political instability, strikes, health, ...



The drawbacks of the *external* network

Modern services are provided by geographically distributed datacenters, but (typical) providers have no control over the external network

➔ Building *global-scale* services is tough

- no bandwidth guarantee because the network is a ***black box*** for most providers (not for Google!)
 - Good for a multi-owned heterogeneous Internet
 - Serious issue for a service provider owning one data center:
 - using inefficient overlay networks, because of the mismatch between physical topology and logical routing
 - trying to infer network properties that typically change dynamically



Geographical replication

If network latency and availability matter → there is a strong motivation for geographically distributing services around the world:

- 1. Improving availability** through *redundancy*, as not all areas are likely to lose power, experience an earthquake or an hurricane, or suffer strikes and social riots at the same time
- 2. Decreasing client-server latency due to speed-of-light “delays”**



If you are a cloud provider or you manage a popular service, one datacenter is insufficient

Multiple Internet Data Centers (and not just one) represent the back-end infrastructure used by cloud providers and largest digital companies to provide agile, elastic, reliable, secure services delivered through the Internet



Inspiration: the historical 4 utilities

[John McCarthy, MIT Centennial, **1961**]: “**Computing may someday be organized as a public utility**”

(1) Water



(2) Electricity



(3) Gas



(4) Phone/Internet



Implementation of the initial vision

[Jeff Bezos, CEO Amazon, **2006**]: “It’s important to sell unused resources of a datacenter so a cost may become a profit”

→ This represents the birth of **CLOUD COMPUTING**

[Steve Ballmer, CEO Microsoft, **2016**]: “The cloud is the future, everything else is accessory”



Two main paradigms

Service

Deployment

Computer science and engineering are not commodities



Service paradigms (3 *prevalent*)

- *Infrastructure as a Service* (IaaS)
- *Platform as a Service* (PaaS)
- *Software as a Service* (SaaS)



Amazon (AWS): IaaS + PaaS

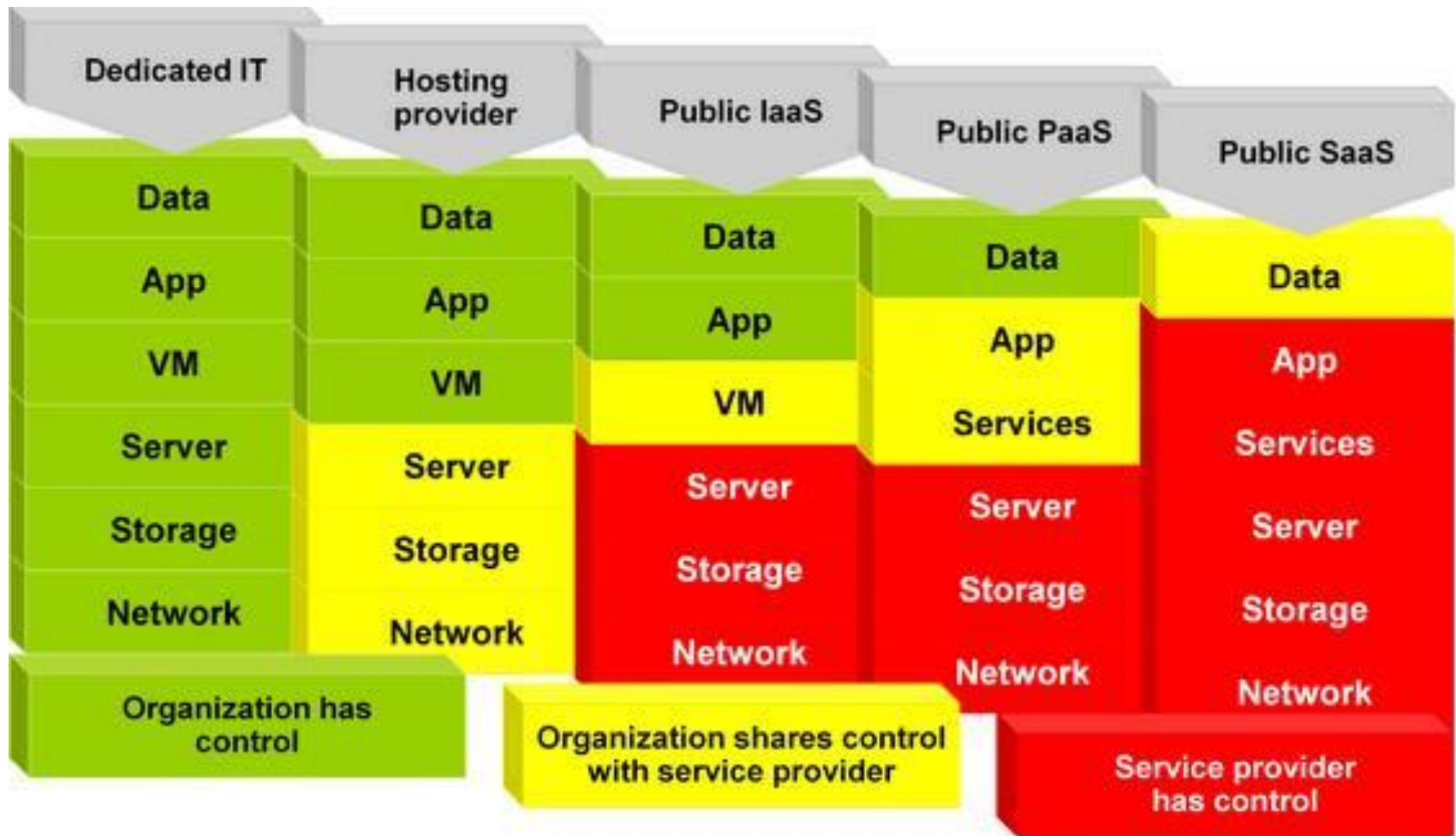
E-Commerce Service (ECS)	Alexa Web Informat. Service (AWIS)	Alexa Web Services Platform (AWSP)	Mechanical Turk (MT)	Historical Pricing (AHP)	Simple Queue Service (SQS)	Alexa Top Sites	Amazon Web Sites Amazon.com Amazon.jp Amazon.fr Amazon.ca Amazon.de Amazon.co.uk joyo.com
AWS – Amazon Web Services Shared Code							
Amazon Technology Platform							

A huge step ahead for Amazon's engineers. This is much more than offering housing or hosing services. **This means sharing the same resources and services of you with your customers**

Selling services. Which ones?

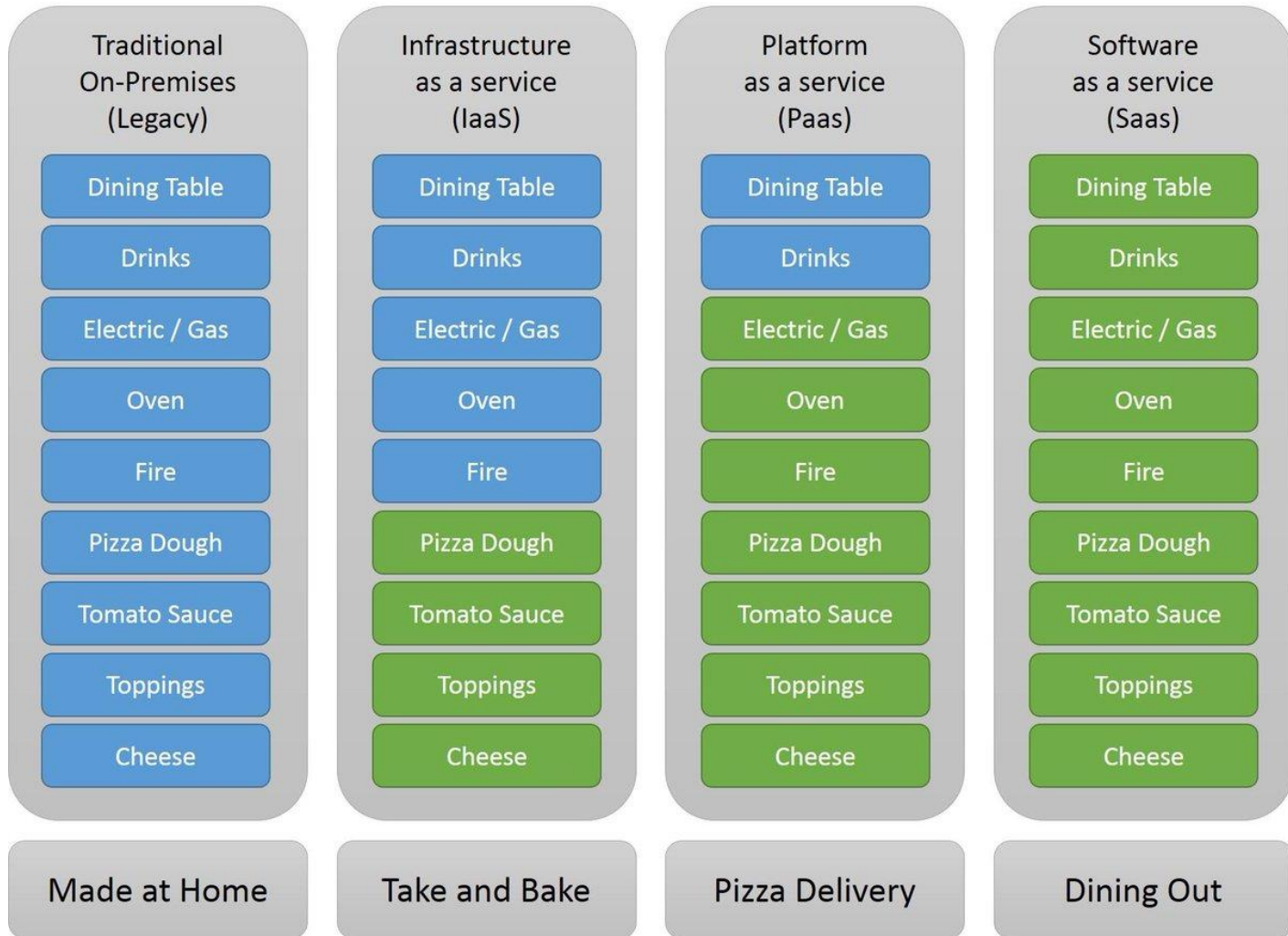


Cloud services: *which organization is responsible?*



Example in a different context

Pizza as a Service



■ You Manage

■ Vendor Manages



Deployment models (2 *prevalent*)

Public cloud

The cloud infrastructure is made available to the general public or a large industry group and is owned by an organization (i.e., **cloud provider**) selling one or more types cloud services

Private cloud

The cloud infrastructure is operated for one organization. It may be managed by

- the organization itself (*private cloud*)
- or a third party (*exclusive cloud*) and may exist on-premise or off-premise



Deployment models

Cloud Provider Infrastructure

Cloud Customer Data Center

① Private



② Community



③ Public



④ Hybrid



Deployment models

1. **Public cloud**
2. **Community cloud**. The cloud infrastructure is shared by several organizations and supports a specific community that has shared concerns (e.g., business mission, security requirements, policy, and compliance considerations). It may be managed by the organizations or by a third party and may exist on premise or off premise
3. **Private cloud**
4. **Hybrid cloud**. The cloud infrastructure is a composition of two or more clouds (private, community, or public) that remain unique entities but are bound together by standardized or proprietary technology that enables data and application portability

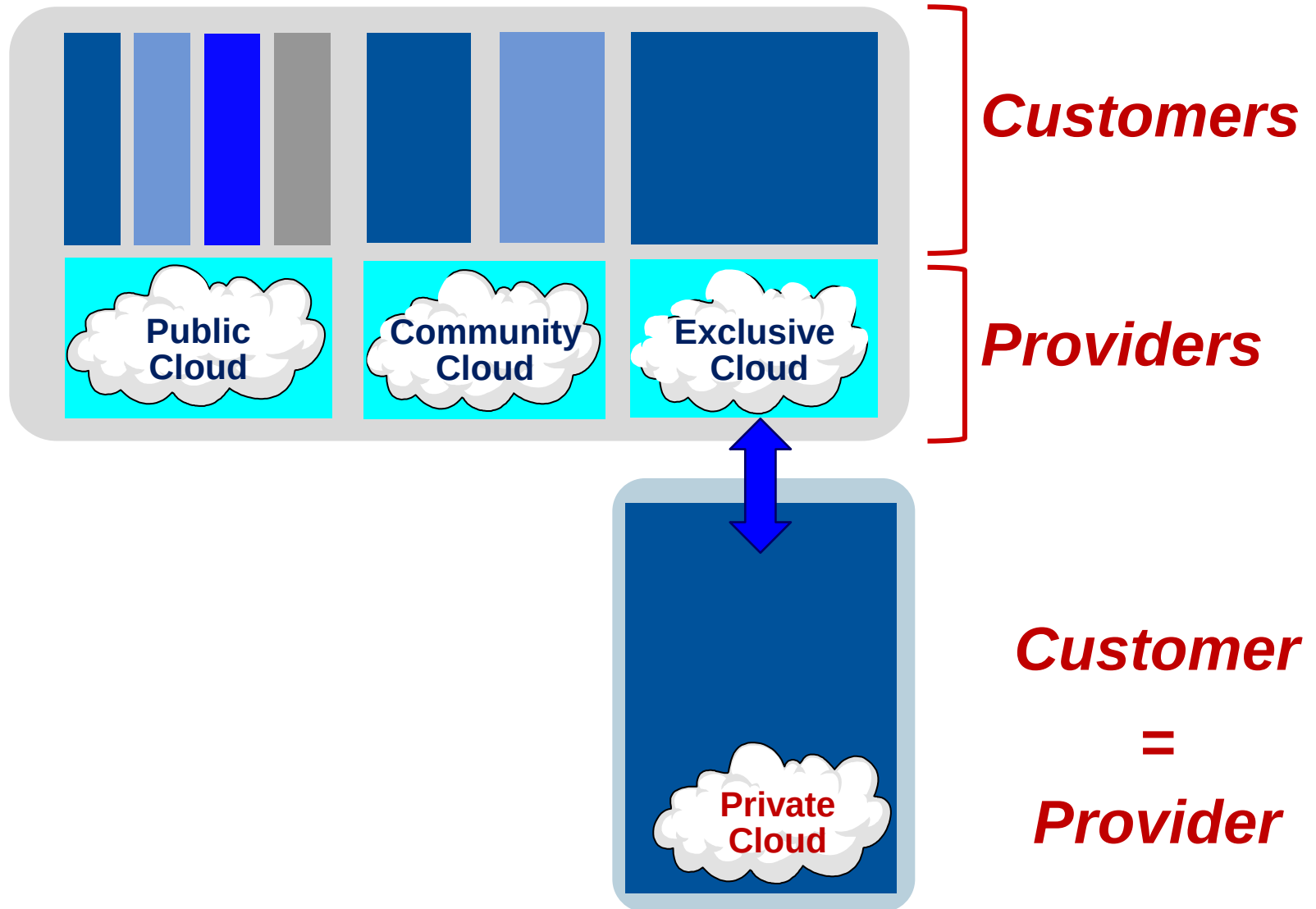


Community clouds - *examples*

- Public Administration
- Health care
- Small-Medium Enterprises
 - Vertical
 - Provider-Clients
- ...

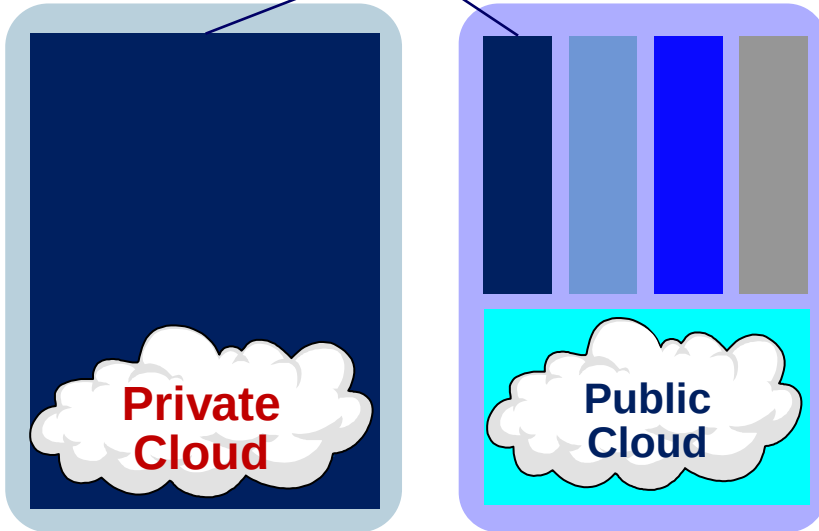


Deployment models

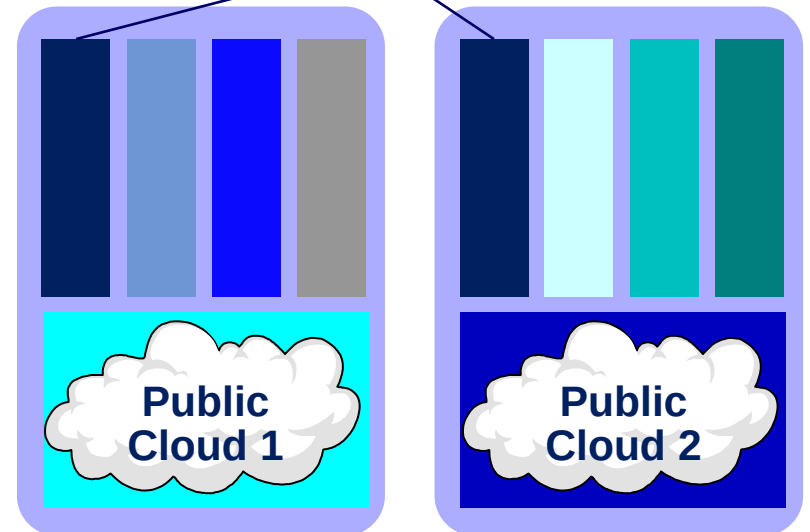


Hybrid solutions

Customer



Customer

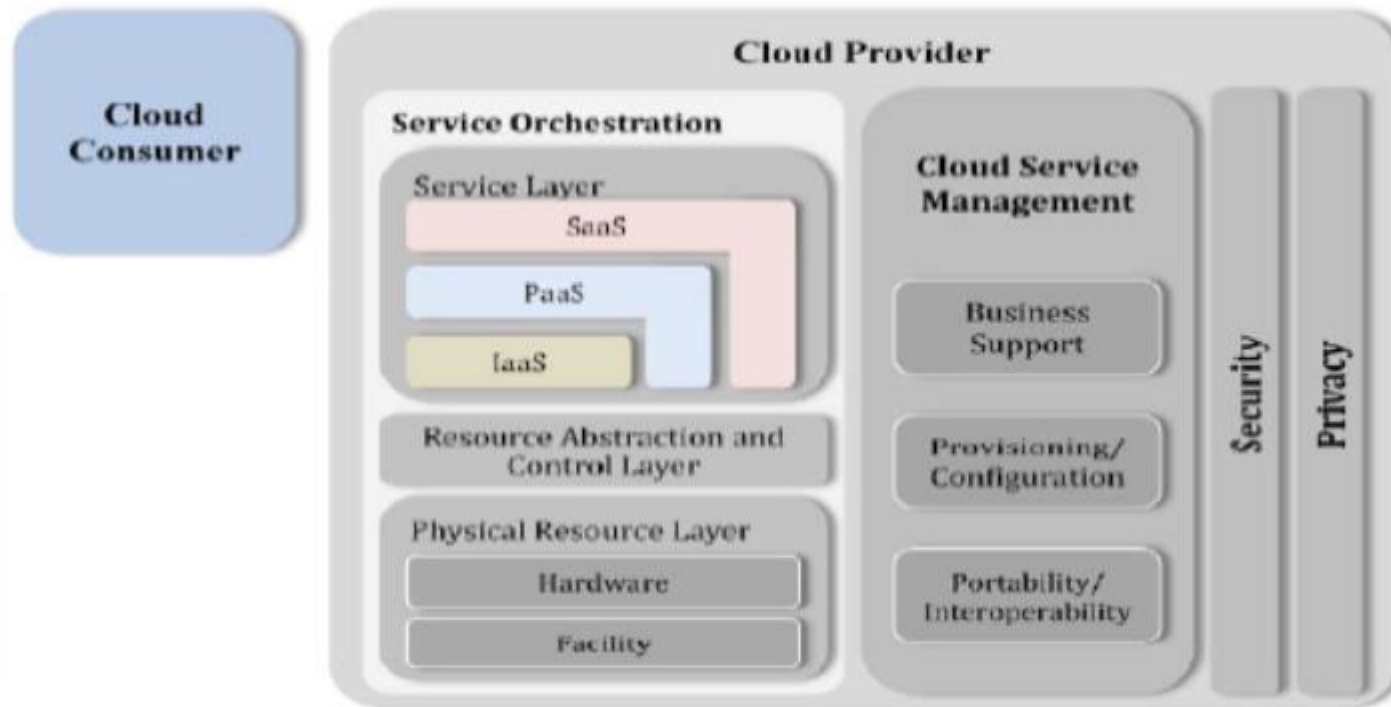


Offered vision of the cloud

1. **A logically isolated set of resources** (disk, memory, processor, network)
2. **Elastic** (*scalability and reverse scalability*)
3. **Reliable** (*always available resources*)
4. **Secure**
5. **Pay-per-use cost model**

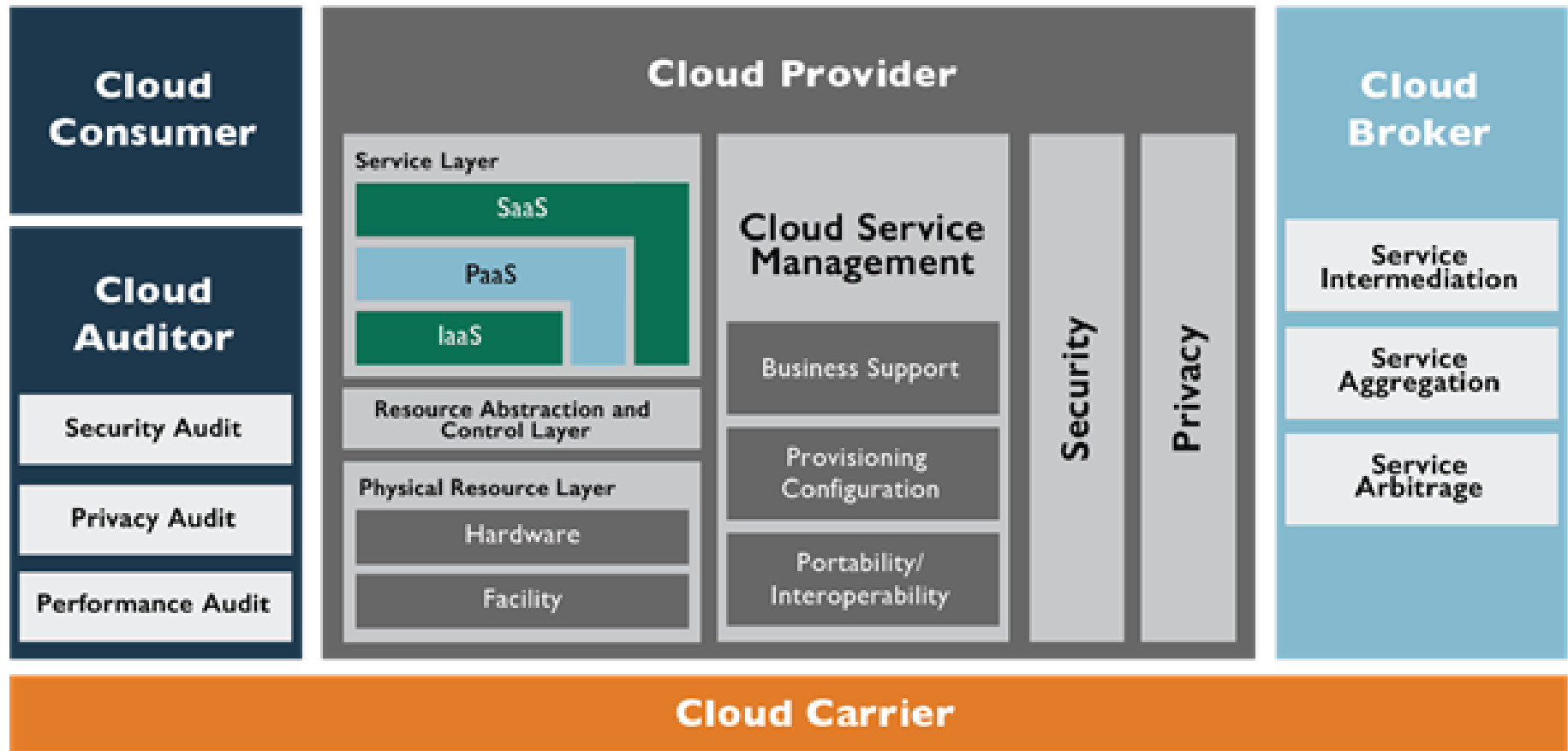


NIST definitions



“Cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a **shared pool of configurable computing resources** (e.g., networks, servers, storage, applications, and services) that can be **rapidly provisioned and released** with minimal management effort or service provider interaction. This Cloud model promotes **availability, elasticity, security** [...].”

NIST definitions: overall picture



Cloud providers



Cloud providers

Only a handful of major IT players can really build a similar massive infrastructure

amazon.com



Microsoft

Windows Azure Platform



Worldwide telco operators are accessing the cloud market



Recently, other big companies

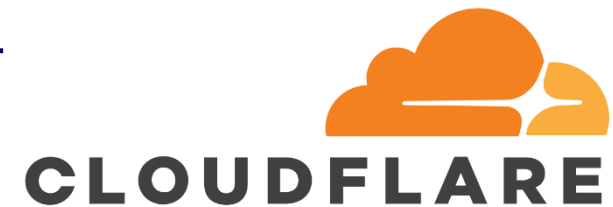


Specialized cloud providers

- **Akamai** → Content delivery (market leader) and security services



- **Cloudflare** → Security (35% DNS, API and SaaS protection, DDoS protection, SSL communications) and Content delivery through *anycast* technology



The big players [Gartner, 2013 and 2014]



The distance augments [Gartner, 2016] (just MS and Google improve, still behind AWS)

Figure 1. Magic Quadrant for Cloud Infrastructure as a Service, Worldwide



Source: Gartner (August 2016)

The top player [Gartner, 2019-2022]

